Final Report

Team Gamma

Romano Almazan (301470828)
Ahmed Badwy (301570091)
Brendan Sng (301560395)
Ryan Thomas (301552334)

Abstract

This report follows team Gamma's journey through developing a dog ball launcher from conception to completion. This report covers topics like the problem our product is meant to tackle, team working which includes developing and sticking to team values and norms, as well as taking a deep dive into the concept generation and elimination process. Starting with the problem, our product is designed to target a certain demographic of people who may be physically limited and unable to play or interact with their dog in that sense. Our product will enable people to play, interact, have fun, strengthen relationships, and build bonds with their dogs by allowing the dog to play a role in operating the machine itself. We used the classic methodology all engineers use when designing new products, empathize, define, ideate, prototype, and test. Since we were unable to get in contact with our client we had to make assumptions in the empathizing stage, we assumed the demographic of people we were designing this product for were people with physical limitations, which could range from accidents both in and out of the workplace or people who have been at a physical disadvantage from birth. In the defining part, we established a perspective on how we would approach this project based on the client's needs. In the ideate phase, we allowed ourselves to have total creative freedom while brainstorming and came up with more than 25 initial concepts. In the prototyping stage, we were able to get a fully functional product despite running into some complications and speed bumps, since at this point we had a physical product in hand we were able to see and come up with some future improvements that we could potentially make. Finally, in the testing phase, we were able to see how our product performs in real-life situations, it was here that we were able to fully assess what we liked about our design and what we could improve on in the future. In this report, we included a discussion, which allowed us to reflect on both the positives and negatives of this journey. One of the topics of discussion was the negative consequences of some of our decisions, in this we described that we severely underestimated a component of our product (the flywheel) and it led us to lose a lot of time due to the lengthy 3D printing times, at the end we had to adapt to the circumstance and find a way ensure the flywheel would work for the time being. The other topic of discussion was steps we would take to improve our product in the future, in this we described that from the start our vision was to have a fully automated product that enabled the dog to fully interact with the product on its own. To do this we brought up the idea of using a sensor and a ball-stopping mechanism to control when the ball was shot, and to this, we would add the necessary code to enable the product to shoot the ball at random time intervals.

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Introduction

This report will be focused on team Gamma's Dog Ball Thrower product and will aim to display the journey it took to build this product from start to finish. From team values, goals, expectations and norms that ensured everyone played their role to the fullest and maximized quality performance of our final product, to problem definition and product vision which was to help people with physical limitations play fetch with their dog amplifying the owner-pet bond and allowing owners to enjoy the company of their pets. This report is designed to take any reader through the entire process of building a product, covering other topics like the concept generating phase where we were able to sort our main ideas into three different groups: exterior, interior, and projection components, and the concept compressing process where as a group we eliminated ideas that we thought didn't fit, was too time-consuming, too costly, or too complex with the vision we had. Along with covering the importance of teamwork and how it has the potential to boost group work. Finally, this report will include a discussion page where we will discuss some difficulties such as budgeting, 3D printing times, lost time due to not hearing back from the client, our limited access to dogs, and also some future improvements we have in mind for this product such as the complete automation of our product and the safety mechanism. This is followed by a conclusion that will consist of results derived from elements throughout the entire paper.

Team Charter

Our team charter consisted of expectations, goals, and norms. Together these expectations, goals, and norms have allowed us to be productive despite the heavy course load we are all enduring. One of our big expectations for this group was that everyone would carry their weight and play their role to maximize the quality of the potential outcome. Another main goal/expectation was communication, we wanted communication between members to be versatile to get as many perspectives as possible and have good organization and time management. A main obstacle that we were able to anticipate was time management and schedule planning because everyone has their schedules and work to tend to. Some of our main norms fall under the category of work, meeting, communication, and consideration norms:

Work Norms:

Our work norms consisted of trying to spend at least 6 hours a week working on this project, understanding each other's strengths and weaknesses and assigning roles based on this to maximize efficiency. taking all options, perspectives, and opinions into consideration to find the best solution, setting goals to organize and schedule ourselves, set due dates for phases in our project for efficient time management

Meeting Norms:

Our meeting norms were to ideally meet once a week for 30 minutes to an hour but meeting times can differ depending on the amount/importance of the work

Communications Norms:

Communication norms consisted of flexible call hours. Discord and Instagram were the primary means of communication

Consideration Norms:

Consideration norms for us were to keep an open mind to all suggestions, options and solutions.

Problem Definition

Product Vision Statement:

Our vision for the product we will be building is to help people who are physically limited to play fetch with their pets, enjoy their pet's company and strengthen the owner-pet bond. With this machine, we hope to improve the life of dog owners and dogs themselves by creating memorable moments and memories between them.

Project Specifications:

Due to us having a very specific vision for our product's present and future, we came up with specifications that we wanted to integrate into our product to help us reach our vision, these being:

- The product must have a safety mechanism in future to prevent injuries to the dog and owner when the ball is launched
- The product must be portable to allow for easy transportation
- Well-balanced, cannot be knocked over easily
- Simple and easy to use for owners and dogs, user-friendly
- It must be modular, to allow for future upgrades and improvements
- The power source must be affordable, easily accessible, and allow for the machine to still be portable (batteries)

Metrics:

In the early stages of our product, we projected some early metrics that acted as goals of what we wanted to achieve through our product. Initially, the preferred weight for our product was between 2kg and no more than 4kg. Currently, the main segment of our product is around 2kgs so we are possibly looking at getting close to 4kgs when the project is 100% completed. We initially planned for our product to be 30-40 cm all around (cube), but as of right now, it's looking like it will be bigger. Our current product is bigger and heavier than we expected it to be, this is only a temporary issue as once we have our product fully functional and working the exact way we visioned we can look to deduce and make adjustments in the portability area. Our goal projectile distance is 20-30 ft, our most recent test saw the ball only shooting around 8-9 feet (rough estimate), this is because our flywheel diameter wasn’t big enough and we had to add material to it to increase the diameter putting the flywheels off balance and lowering the power, velocity, and accuracy with which the ball was shot at. Our solution to this was very easy as we just reprinted bigger flywheels, and this also fixed the centring problem we had.

Limitations:

Some limitations we ran into were budgeting, it was hard to find the parts we needed at a reasonable price without cheapening out on quality and performance. Along with this came the time it took for the parts to ship since we could only find them online, this pushed back our full assembly for a bit. We were also limited by 3D printing times as many of our pieces took days (one took 3) to print to achieve our desired quality. Because none of the people in our group own dogs, our testing time and accessibility to dogs were very limited, though we had connections to people with dogs it meant that we would have to abide by their schedules in turn scrambling and forcing us to adjust our schedules. We didn't have access to lots of equipment needed to build this prototype minus a 3d printer, like before we had connections to people with the equipment that we needed but again it meant working on their schedule. Due to not hearing back from our client, we lost about 3 weeks of crucial time putting us in a bit of a time crunch.

Concept Generation

In the concept generation phase, we assorted our ideas and put them in three separate groups:

Exterior Components:

1. Steele base, to lower the center of mass and make our prototype stable and hard to knock over
2. Most components should be made out of carbon fiber to keep the rest of the prototype light
3. Cube design with rounded edges makes it easy to store for people who may not have a lot of space while also eliminating the danger of sharp corners
4. The cosmetic idea, most other dog ball throwers on the market look like toilets, we will configure ours to avoid this
5. Make the prototype modular to allow for future improvements, attachments, and fixes.
6. Flashlight/bright light source for the dark
7. Camera to monitor dog, the camera can also be configured to assist other mechanisms in the product like the safety mechanism

Interior Components:

8. Funnel, to allow and make it easy for the dog to use the machine by itself
9. Automation, we plan to add code to make the product fully run by itself
10. The safety mechanism incorporates a sensor that detects anything within 1-2 ft of the nozzle and locks the entire machine if something is present
11. Mechanism to stop the ball as soon as it's dropped into the funnel. Prevents the ball from shooting out instantly and causing injury, it could be configured to release the ball at random intervals making for a more exciting experience
12. Conveyer belt, so the ball doesn't hit the flywheel as soon as it dropped
13. Slope, allowing for the ball to roll into the flywheels with some speed and velocity already built up
14. Battery powered (simplest method), using 2 8-pack battery holders will provide enough voltage for the motors to spin at a strong speed, also batteries are easily replaceable
15. Rechargeable (user-friendly method), since replacing batteries over time can add up making the product rechargeable will be ideal

Projection Components:

16. Magnetic plates to shoot the ball
17. 2 flywheel mechanisms to shoot the ball
18. Some sort of air pressure-related mechanism to shoot the ball (similar to shirt cannons)
19. An adjustable incline to complement the flywheel mechanism allows the ball to be shot at different angles resulting in different heights, projectiles, and distances

Concept Screening

Ideas Cut Out: 1, 2, 12, 16, 18
Ideas Used for Concept Matrix: 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15, 17, 19
Concept Matrix:

	Cube case	No toilet shape	Modular prototype	Lights	Built-in Camera	Funnel	Automation	Safety Mechanism	Ball Stopper
Cost	0	0	+1	-1	-1	-1	-1	-1	0
Durability	0	0	0	0	0	0	0	0	0
Lightweight	0	0	+1	0	-1	-1	0	0	0
Portable	0	0	0	0	0	0	+1	0	0
Manageable	0	0	-1	0	+1	0	+1	+1	+1
Net Score	0	0	+1	-1	-1	-2	+1	0	+1
Rank	2	2	1	7	6	3	6	4	4
Status	**Continue**	**Continue**	**Continue**	**Reject**	**Reject**	**Revise**	**Revise**	**Revise**	**Revise**

continued	Slope	Battery Power	Rechargeable Battery	Dual Flywheel	Adjustable Incline
Cost	+1	0	-1	0	-1
Durability	0	0	0	0	0
Lightweight	0	0	0	0	0
Portable	0	0	0	0	0
Manageable	0	0	0	0	0
Net Score	+1	0	-1	0	-1
Rank	2	2	5	1	3
Status	**Continue**	**Continue**	**Revise**	**Continue**	**Revise**

Rank Legend:

Lower Number (ie. 1,2,3): More important
Higher Number (ie. 5,8,9): Less important

Presentation of Top Ideas

Our top ideas for this product are as listed:

Flywheel:

The flywheel was our first and top concept shooting mechanism from the very beginning of the semester since it is easily understandable, not complex(in case we need to make adjustments), and has the potential for great speed and torque. The simplicity of the flywheel mechanism proved to be beneficial to us as we had to adjust the diameter of the flywheels multiple times throughout this project. This mechanism is essentially 2 wheels attached to motors on either side of the tube, one spinning clockwise and the other spinning counter clockwise, once the ball makes contact with the flywheels and pinches in between them the torque and rpm of the wheels are spinning at allow the flywheels to launch a ball at high speed and velocity.

Funnel:

One of the conceptual components that made our project stand out in our eyes is the funnel. Its main purpose was to allow the dog to put the ball into the machine by itself which in turn would lead to us having the option to completely automate the machine in the future so a dog could safely and independently use it. Incorporating the funnel would have led to us implementing a sensor and a stoppage mechanism so the ball could be shot at random intervals. In the end, though we had complete models ready to print, due to limitations with time, material, and budget, the funnel proved to be too costly in those terms and we ended up not using the funnel although we plan to integrate it in the future.

Automation:

Automation is currently a completely conceptual concept for us at the moment, we would like to include it in our product but due to the lack of time, we had to let go of the idea for the time being. Our plan was stated previously in that we would have liked to have a sensor and something connected to the sensor that could stop the ball, then we would implement code that would release the ball at random intervals allowing it to hit the flywheel and be unpredictable so the dog won't get into a routine and still keep it excited to use the machine. This will also help in getting the dog to use the machine by itself.

Safety Mechanism:

Along with the automation comes a safety mechanism to prevent injury to the dog or the owner. It will include a sensor and code that will turn off/ lock the shooting

mechanism if an object is detected within a foot or less of the nozzle. We aim to include this along with automation.

Concept Selection

Matrix of Top Ideas:

Concepts

			Casing	Modular	Prototype	Power	Source	Safety &	Automation
	Weight	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**	**Rating**	**Weighted Score**
Cost	17%	1	0.17	2	0.34	1	0.17	3	0.51
Easy to Use	5%	2	0.10	4	0.20	2	0.10	4	0.20
Safety	16%	1	0.16	2	0.32	2	0.32	1	0.16
Durability	7%	1	0.07	3	0.21	3	0.21	2	0.14
Lightweight	10%	3	0.30	2	0.20	2	0.20	3	0.30
Portable	10%	3	0.30	2	0.20	3	0.30	2	0.20
Number of Parts	7%	3	0.21	1	0.07	4	0.28	3	0.21
Reliability	6%	2	0.12	3	0.18	4	0.24	1	0.06
Time	12%	2	0.24	1	0.12	5	0.60	4	0.48
Aesthetics	5%	5	0.25	5	0.25	5	0.25	5	0.25
Manageable	5%	3	0.15	3	0.15	3	0.15	4	0.20
Total Score			2.07		2.24		2.82		2.71
Ranking			2		1	1		3	

Continued			
		Additional	Parts
	Weight	**Rating**	**Weighted Score**

Legend:

Lower Number (ie. 1,2,3): More important
Higher Number (ie. 5,8,9): Less important

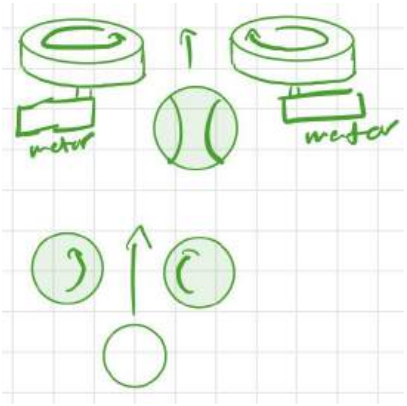
Cost	17%	1	0.17
Easy to Use	5%	2	0.10
Safety	16%	2	0.32
Durability	7%	3	0.21
Lightweight	10%	3	0.30
Portable	10%	2	0.20
Number of Parts	7%	4	0.28
Reliability	6%	3	0.18
Time	12%	4	0.48
Aesthetics	5%	5	0.25
Manageable	5%	4	0.20
Total Score			2.69
Ranking			4

Justification:

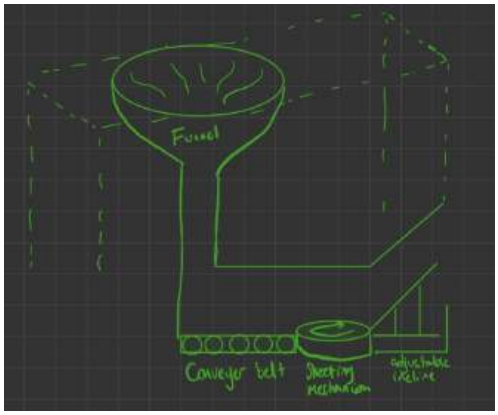
Cost is rated the highest because of the $100 budget that we are working around. Time and Safety are the next two highest rated. We needed to make our product safe for us to test and for dogs to use while completing all this in an orderly fashion to stay on time with our due date (the end of the semester). Lightweight and portability are next in line despite being in our vision statement, if we can't make a product that is safe and completed on time then how can we refine the product and get to meet our goals? The remaining six items are rated less than 10% despite these being of importance. They are not as important and we do not have to focus a lot of our energy on some of these things.

Detailed Design

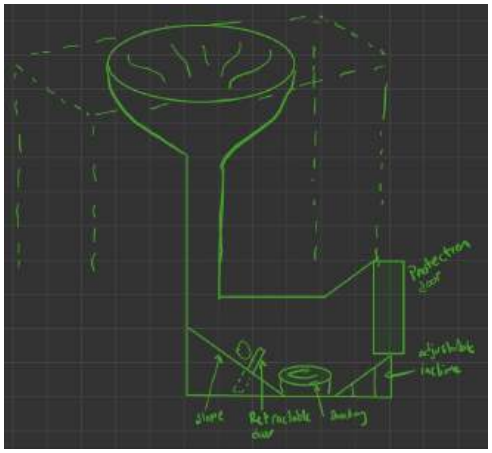
Schematics:



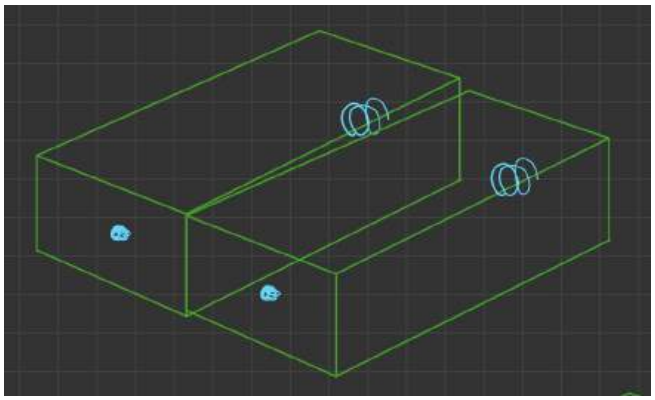
First Flywheel concept



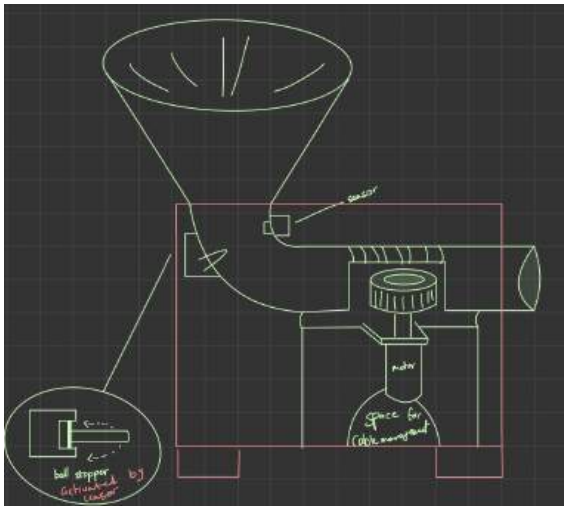
Concept with conveyor belt



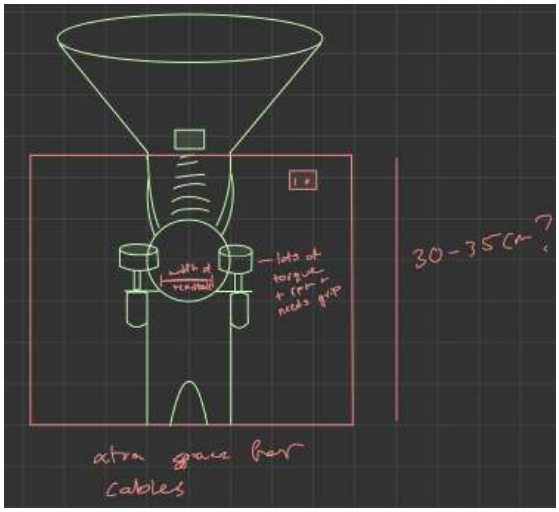
The concept with slope and stopper



Battery pack schematic



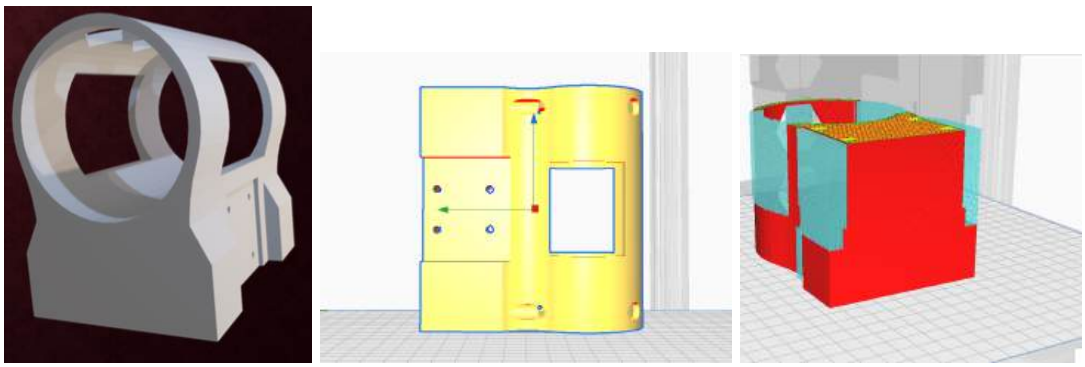
Final schematic side view



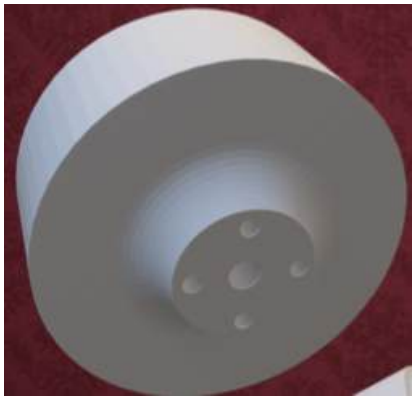
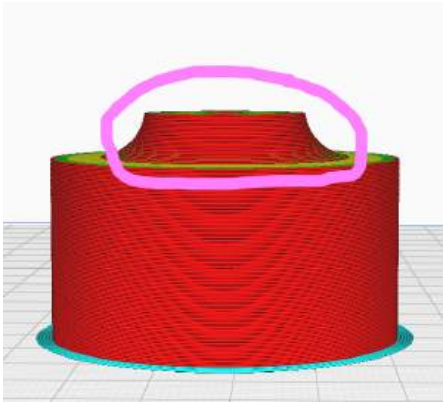
Final schematic front view

3D Models/Slicing:

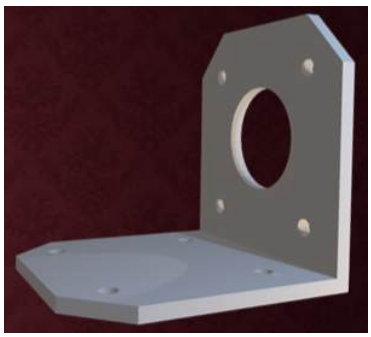
3D models/slicing of the funnel



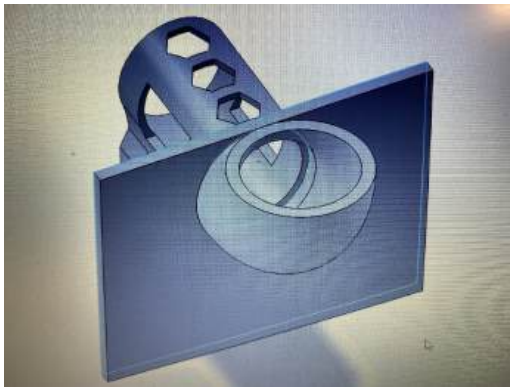
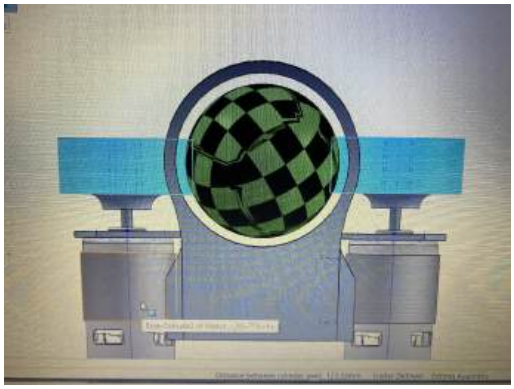
3D models/slicing of the shooting tube

3d model/slicing of the flywheels



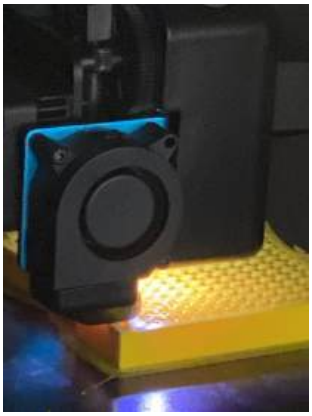
3D model of motor and flywheel 3D model of motor 3D model of the L bracket

3D model of insert tube 3D model of everything put together

Final product:

The initial fitting test



Printing process

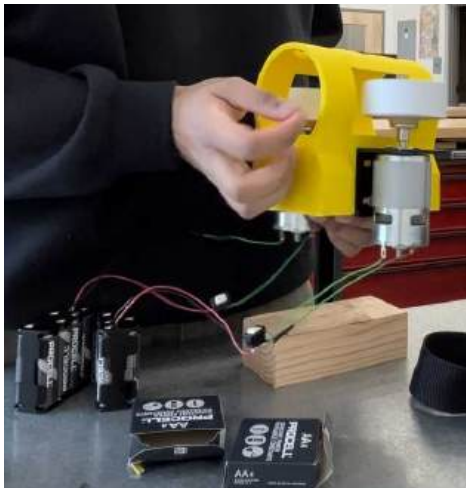
Initial flywheels

Flywheel and coupling

Final flywheels and motors

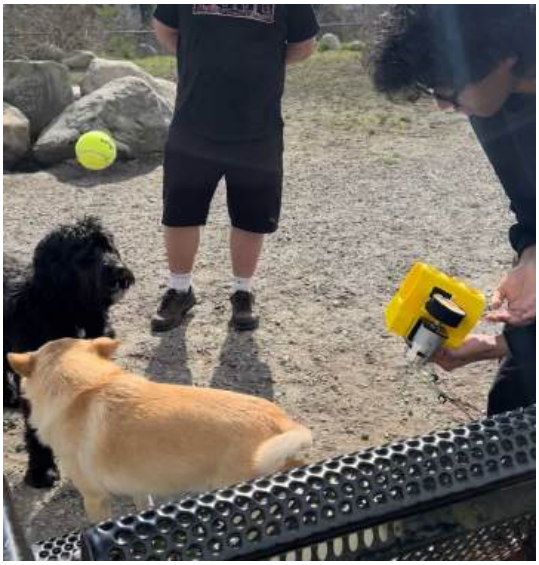
Partially assembled tube

Fitting test



Fully assembled tube with electrical components

Testing with dogs:



Teamworking

As we are near the end of this project we were able to come up with a final version of our project management report:

Get Project Specifications

Before even starting a project having the specs is especially important. From a customer service perspective if you give the customer.

People Involved:

Brendan

Generate Concept Solutions/Designing

Before building our project we have to generate some concepts. The main idea is to think of problems we could encounter and find solutions to the problems. For example, we need to throw a ball well how? During this process, we will generate possible solutions to this problem.

People Involved:

Brendan, Romano, Ahmed, and Ryan

Sketch the Final Solution

After generating some concepts we have to narrow them down and find one solution. This could be a combination of various components from our concepts or could be something entirely different. The sketches should include drawings and schematics of the final product and every component within, with a listing of specifics and notes. The final sketch will start to give us a direction on where to go and lead us to start the manufacturing process.

People Involved:

Ryan

Research Materials

To build something you have to have the correct materials. Researching materials is crucial especially if you are on a budget. For example, something like carbon fiber can get expensive because you have to buy things like the fabric, epoxy and other minor things needed. We also have to find out which materials will be the best fit for our final product. This means if a material is cheap but is super fragile to work with it is probably not a good idea to use this material to make our prototype.

People Involved:

Brendan, Ryan

CAD Drawings

Using school-provided software, Solidworks. We will design prototypes using this to have a formal blueprint of how we wish our product will look.

People involved:

Romano, Ahmed

Manufacturing

We need to build our product. This means booking shop times and getting access to special equipment. Getting access to resources. Managing resources and materials under budget. Quality assurance.

People Involved:

Brendan, Romano, Ahmed, Ryan

Inspection

This is to make sure that components are up to par and address problems that may cause a disaster and fix them. This also makes sure that there were no flaws during the manufacturing process like any cracks in parts

People Involved:

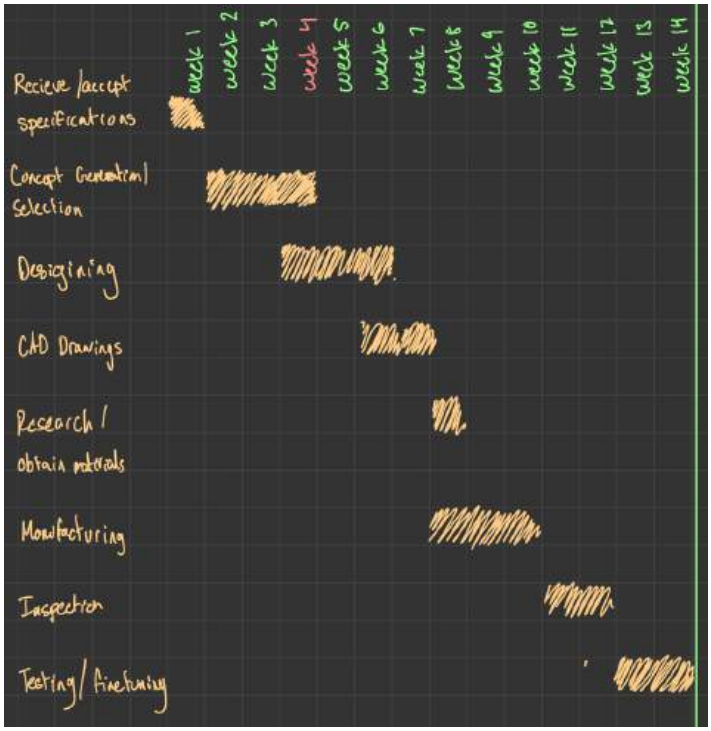
Ryan, Brendan, and Romano

Testing

This is to make sure the product works and meets our standards and to ensure that it functions properly in real life scenarios rather than theoretical scenarios because theoretical results may differ from the actual result (ie. tested with real dogs at a dog park)

People involved:

Ryan,



In summary, our roles can be summarized as:

- Brendan Sng: Resources, finance, communications, quality control/manufacturing (3D printing)
- Ryan Thomas: Calculations, mechanical, electrical, assembly, parts, testing
- Romano Almazan: Solidworks/modelling, schematics, parts
- Ahmed Badawy: Solidworks/modelling, schematics

We believe that as a team we were able to distribute all the work necessary for this project fairly while also making use of everyone's strengths and circumventing weaknesses. For example, Romano and Ahmed had already completed MSE 100, they were equipped with the required Solidworks skills needed to come up with 3D models that we could use to print and were, therefore, in charge of the main modeling work. Brendan owns and manages his band, and due to his experience in managing groups, we elected him as our product manager. Ryan has plenty of experience in mechanics and electronics as a result of building four go-karts in high school, l, as a result, he took the main lead in the assembly and testing of the product.

One challenge at the start for us was communication and meeting up. Due to the stress and workload of other courses, we were all taking and especially as midterms rolled around we found that when we did do group calls on Discord not everyone was able to make it. We solved this problem by assigning specific days and times to meet at the school computer lab and work together there. This is how we found our stride and we noticed that we were way more efficient and productive while all working in person, if one of us had a problem or needed help all of us were there to support. Working in person also made it easier to share ideas, and opinions, and to deduce problems to lead to a solution.

Discussion

What were the negative consequences of some of your decisions?

One of the main negative consequences as a result of decision-making for us was the flywheel process and specifically the wheels themselves. The flywheels were an integral part of our prototype as it was the shooting mechanism we had chosen. On average, each pair of flywheels we printed took around 6-8 hours plus adding the grip tape took another 12 hours or so as we wanted to ensure the glue set properly. It should be noted that the flywheels are still usable while the glue is setting as long as the grip tape is secured with rubber bands. During the entire process, we severely underestimated the required diameter of the flywheels and had to reprint them twice (three times in total), due to the printing and waiting for the grip tape to set, repeating this process twice made us lose a lot of time and ground. This especially held back our testing and eventually limited the capabilities of our product on the testing day. We had originally anticipated that our original flywheels (first print) would be too small and not make contact with the ball so we printed a set of larger ones, as we were in the final stages of assembly we noticed that the second flywheels we printed would be too small. While we did start printing bigger wheels immediately, due to personal locations we were unable to deliver them to the group member assembling the product. To cope with this we took some thick plastic sheets and wrapped them around our current flywheels to increase the diameter. While this solved the diameter problem a new problem arose, because we added extra material to the flywheels the center of mass was off leading it to be unbalanced when spinning and causing vibration. This in turn reduced the power and how far our ball could go, but it was able to shoot the ball and we were able to test it. We feel that if we didn't underestimate the diameter we could have used that time to touch up other aspects of our product or do more tests before the presentation but since we currently have the corrected flywheels ready to go we look forward to testing our product at its full potential and capacity.

If you have more money, time and/or engineering experience, how would you improve your design?

As stated before, if our budget, time, and engineering experience were increased we would have loved to try to fully automate our product. Since our main assumption and vision for this product is to be of aid to people with disabilities or people who have ill health, completely automating our machine would be of great benefit to our target consumers. We would first start by incorporating the funnel back into our prototype as it allows dogs to be trained to put the ball into the funnel so the machine can do the rest of the work.

Next, as seen in our final design we would incorporate a sensor and a stopping mechanism. The sensor would be any type of object detection sensor that would fit our budget and our quality standards. The stopping mechanism as seen in our final design would be a small stick-like object powered by a small motor that can rotate 180°, when rotated to 90° it will stop the ball, and when rotated to either 180° or 0° the ball will fall through. The stopping mechanism will be connected to the sensor and will be coded to activate every time the sensor senses the ball. Once the sensor senses a ball and the stopping mechanism catches it we will come up with code to make the ball shoot out at random intervals. We can do this by coding the stopping

mechanism to deactivate after random amounts of time have passed, for example, in one turn the ball will take 3 seconds to shoot and the next it might take 7. The vision we have for implementing this into our product is solely for the dog's enjoyment, as we don’t want our machine to be predictable in the sense that the dog will know exactly when the ball shoots and eventually get bored. We want to keep our product exciting so that for the dog, no matter how many times it plays with our product it will always feel like the first time.

Conclusion

This report captures and outlines team Gamma's journey and use of teamwork in designing and building a dog ball thrower prototype, from the team values and norms, problem definition, and problem vision to the concept generation and compression, to the building process and its strong points and limitations.

Our team's vision for the product was to help people who may be physically limited to play fetch with their pets, allowing them to enjoy the company of their pets and strengthen their owner-pet bond.

The report concludes with a discussion of some of the challenges our team faced, how we plan to improve our product in the future and all the ideas we have to do so. Though our team had limitations in budgeting, 3D printing times, and limited access to dogs, we were able to create a product that works efficiently through our strive for success, our commitment to teamwork, and finding our stride with communications.

Testing links:

Test 1: https://www.youtube.com/shorts/wd-qJUAzYRY

Test 2: https://www.youtube.com/shorts/Kr0xP0e3ssY